NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

NTU Academy for Professional and Continuing Education

(SCTP) Advanced Professional Certificate

Data Science and AI

Coding Class Norms

For a Smooth & Productive Session

- **Maintaining Class Flow**
  - Complete all pre-work — class builds on it.
  - If pre-work isn’t done: listen, observe, catch up later.
  - If majority face the same issue, we will pause to address together.
- **Self Sufficiency**
  - Review instructions/materials.
  - Search online — Official docs, Google Search, Stack Overflow, AI assistants.
  - Attempt troubleshooting
- **Peer Learning**
  - Build relationship, collaborate with your buddy or assigned group to solve problems.
  - Supporting peers strengthens teamwork, deepens understanding,
  - Enhance learning from diverse skills and viewpoints

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NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

1.2 Introduction to Databases

Module Overview

1.1 Introduction to Data Science

1.2 Introduction to Database

1.3 SQL Basic - DDL

1.4 SQL Basic - DML

1.5 SQL Advanced

1.6 Introduction to Numpy

1.7 Introduction to Pandas

1.8 Exploratory Data Analysis (EDA) Basic

1.9 EDA Advanced

1.10 Data Visualisation

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Agenda

1. The Data Landscape - Brief History of Databases
2. Blueprinting: ERD Construction & Keys (PK/FK)
3. Normalization (1NF-3NF)
4. Wrap Up & Exit Ticket

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A Brief History of Databases

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Workshop

https://github.com/su-ntu-ctp/6m-data-1.2-intro-database/blob/main/lesson.md

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